



Declining Energy Return on Investment of Oil and Its Implications for the Long-Term Sustainability of Energy Systems



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Abstract: The long-term viability of fossil-based energy systems is closely linked to the net energy they deliver to society. However, many widely cited estimates of oil energy return on investment (EROI) rely on data that no longer reflect current production structures. In particular, the rapid expansion of shale oil in the US has fundamentally altered the relationship between production costs, market prices, and net energy yields. This study revisits recent EROI values for oil in the US and Russia by examining their relationship with production costs and observed price dynamics, together with projected trends in EROI decline. A complementary assessment based on monetary return on investment (MROI) is also conducted to capture the economic dimension of energy extraction. To place these findings in a broader context, the analysis considers long-term changes in the gold-to-oil price ratio and compares the energy content of crude oil with the mechanical output of human labor. The results indicate that effective net energy returns from oil continue to decline, with implications that extend beyond production economics. In the case of the US, the analysis points to an emerging net energy deficit when recent production structures are taken into account. These developments suggest increasing constraints on the capacity of oil to support sustained economic activity. The findings underline the need to reassess the role of oil within future energy systems, particularly in light of growing concerns regarding resource limits and long-term sustainability.

Keywords: Oil energy return on investment; Net energy; Monetary return on investment; Energy sustainability; Resource constraints; Oil production economics

1. Introduction

Oil remains the backbone of industrial civilization, serving as a critical input in approximately 90% of all industrially manufactured products and accounting for around 30% of global primary energy consumption. Its central role has far-reaching economic, environmental, and geopolitical implications. During the 2010s, the shale revolution, driven by advances in extracting unconventional oil resources, particularly in the US, led to a substantial increase in oil production, reshaping global energy markets and geopolitical dynamics. Given these developments, it is essential to evaluate the net energy yield of oil, defined as the energy extracted relative to the energy expended in its production and distribution. In the context of an accelerating global transition toward low-carbon energy systems, the issue of net energy from oil liquids deserves heightened scholarly and policy attention.

The overall decline in the energy return on investment (EROI) for US oil extraction indicates that the depletion of oil fields has increased the energy costs of extraction. Estimates of declining EROI for US oil production are provided in the study (Cleveland, 2005). Gagnon et al. (2009) also present a preliminary assessment of EROI for the world's most critical fuels, oil and natural gas, using time series data on global production and estimated energy inputs. These inputs are derived from monetary expenditures of publicly traded oil and gas companies, combined with estimates of the energy intensities associated with those expenditures.

Limited research has systematically examined, modeled, defined, and empirically tested the relationship between oil prices and the EROI. The studies (Heun & de Wit, 2012; King & Hall, 2011) establish and analyze

the link between oil production costs, market prices, and EROI, highlighting how these factors are interrelated in energy economics. The study (Guay-Boutet & Dufour, 2024) investigates the relationships between net energy ratios (NERs) and key economic variables, namely, market price, production cost, and price-to-cost ratios, for Canadian oil sands extracted through open-pit mining from 1997 to 2016. A simplified econometric model is developed to estimate the correlation between standard EROI and price, production cost, and the price-to-cost ratio of both diluted bitumen and synthetic crude over a 20-year period.

Guilford et al. (2011) describe two general patterns in the relationship between energy gains and energy costs: EROI for oil has been declining slowly but steadily over a long period, and it exhibits an inverse relationship with drilling effort. Energy intensity data for US oil production are also presented. Lübbers & Bredemeier (2019) establish relationships between commodity prices and EROI for oil.

Hall et al. (2014) provide comprehensive EROI estimates for multiple fuel types, analyzing the downward trend in oil EROI and discussing the associated societal impacts. The studies (Cleveland & O'Connor, 2011; Huang et al., 2020) analyze multiple aspects of oil shale, including EROI, energy investment, and the costs associated with shale oil production.

The studies (Carayannis et al., 2020; Safronov & Sokolov, 2014) present data on Russian oil and gas production, production conditions, and EROI for hydrocarbons (both oil and gas) across the five largest oil companies in Russia. According to the study (Safronov & Sokolov, 2014), the EROI for oil production in Russia varied among companies, ranging from 22:1 to 35:1 in 2012. The EROI for light petroleum products during the same year ranged from 5:1 to 13:1. Grandell et al. (2011) provide data on Norwegian oil and gas from 1991 to 2008. The studies (Celi et al., 2018; Celi, 2021) outline methodologies for calculating corporate-level EROI, including estimates for major oil companies.

King (2010) compares two metrics of energy quality, EROI and Energy Intensity Ratio, in the context of fossil fuel consumption and production in the US. Weißbach et al. (2013) present EROI values for oil, gas, and various renewable energy sources, along with monetary return on investment (MROI) and energy intensity data for selected energy carriers. Gupta & Hall (2011), in contrast, provide a comprehensive EROI review covering 12 different energy sources. The study (Salehi et al., 2020) demonstrates that EROI values can be overestimated by up to 75% if embodied energy costs are not considered. By comparing aggregated EROI estimates based on state variables (such as price, energy, and exergy) with disaggregated estimates using process-based properties (specifically exergy cost), the study quantifies the error introduced by aggregation. Dupont et al. (2021) estimate the NER, a more comprehensive indicator than EROI, by accounting for the energy embodied in both intermediate and capital consumption across the entire economy.

Declining production from conventional oil resources has prompted a global shift toward unconventional sources, such as tar sands and shale oil. These unconventional resources are generally more difficult and energy-intensive to extract, resulting in significantly lower EROI values compared to conventional oil. In the study (Murphy et al., 2022), the authors present both a comprehensive literature review and a harmonization of EROI values to enable accurate comparisons across thermal fuels and electricity-generating technologies. Crucially, they advocate for the adoption of point-of-use EROI metrics, as opposed to point-of-extraction values, emphasizing that the energy costs associated with processing, refining, and transporting thermal fuels significantly reduce their effective EROI. For instance, the point-of-use EROI for petroleum is reported to fall notably below ten. Murphy (2014) concludes that, as the aggregate EROI of oil continues to decline, sustaining long-term economic growth will become increasingly challenging and will entail rising financial, energy, and environmental costs. Ecclesia et al. (2022) conduct a case study to estimate the EROI for Portugal and reported a significantly lower value, approximately 3, compared to existing estimates. This value remains relatively stable over an extended period (1960–2014), suggesting that EROI may be largely independent of economic growth. However, when calculated at the final consumption stage using conventional methodologies, the results indicate a declining trend in Portugal's societal EROI. Brandt (2017) presents a mathematical model for analyzing the effects of energy resource depletion across all sectors of the economy and society as a whole, and the study (Brandt et al., 2015) provides the first engineering-based assessment of the energy intensity associated with Bakken (North Dakota) crude oil production and calculates the resulting NER.

The study (Illig & Schindler, 2016) uses dynamic production function identities and an empirical oil price model based solely on extraction data to analyze oil price dynamics during the contraction phase of extraction, exploring implications for various common scenarios. Kleinberg et al. (2018) examine key geological, geographical, and temporal factors affecting tight oil production, along with its inelasticity and broader economic and political impacts.

The study (Delannoy et al., 2021) integrates standard EROI estimates and dynamic functions into the GlobalShift bottom-up model at a global scale. The findings indicate that the energy required to produce natural gas, including both direct and indirect energy and material inputs, currently accounts for approximately 6.7% of gross energy output. This proportion is increasing exponentially and is projected to reach 23.7% by 2050, implying a substantial decline in EROI for gas, i.e., hydrocarbons in general.

Hamilton (2024), a researcher at the University of Edinburgh and founder of Zero Emission Scotland, along

with his colleagues, conducts self-funded research into worldwide oil depletion. Their findings indicate a critical timeline: global access to economically viable oil is projected to decline by the 2030s. Hamilton (2024) calculates the EROI for oil extraction using a thermodynamic model and oil temperature measurements in North Sea wells and found that, although residual oil deposits will persist, the energy required to extract these lower-quality reserves will exceed the energy obtained from their combustion. This phenomenon is attributed to the need to access increasingly deeper and more challenging geological formations. Specifically, their analysis forecasts that North Sea oil will become commercially unviable by 2031, with Norwegian oil reserves following closely by 2032. Similar depletion trends are anticipated globally throughout the 2030s.

Energy intensity is a key indicator of energy efficiency. Aydin et al. (2025) examine the impact of energy security risk, natural resources, technological regulation, and economic growth on energy intensity in the USA from 1990 to 2019. Bajan et al. (2021) identify an inverse pattern between development and energy returns: least-developed regions exhibit the highest EROI yet still fall short of local food needs, while highly developed regions maintain food self-sufficiency despite low EROI through intensive energy use. Continued declines in oil EROI therefore threaten to reduce food production and increase prices. Dorić et al. (2025) introduce a new internal combustion engine concept, with reported efficiencies exceeding 40%. Data from studies (BP, 2022; Cembalest, 2023; Energy Institute, 2024; Eurostat, 2015; IEA, 2024; Statista, 2015; US EIA, 2011; US EIA, 2025) provide key energy parameters, particularly for oil and gas, essential for this research. Additionally, this paper utilizes global population and GDP data from the study (World Bank, 2025).

The primary motivation of this study is to estimate recent EROI values for oil in the US and Russia and to quantify the increasing net energy deficit associated with the ongoing decline in EROI in the US. This study estimates the 2023 EROI for Russian crude oil and, for the first time, derives its corresponding MROI. For the US, the 2023 oil EROI is estimated using two complementary methodologies: (1) projecting the historical decline in conventional oil EROI while adjusting for recent changes in the domestic production portfolio, and (2) deriving EROI from economic parameters by comparing upstream oil production costs with corresponding market prices and the US oil MROI.

2. Net Energy and Energy Return on Investment of Oil

In the context of oil extraction and oil wells, the energy flow exhibits distinct characteristics, as illustrated in Figure 1.

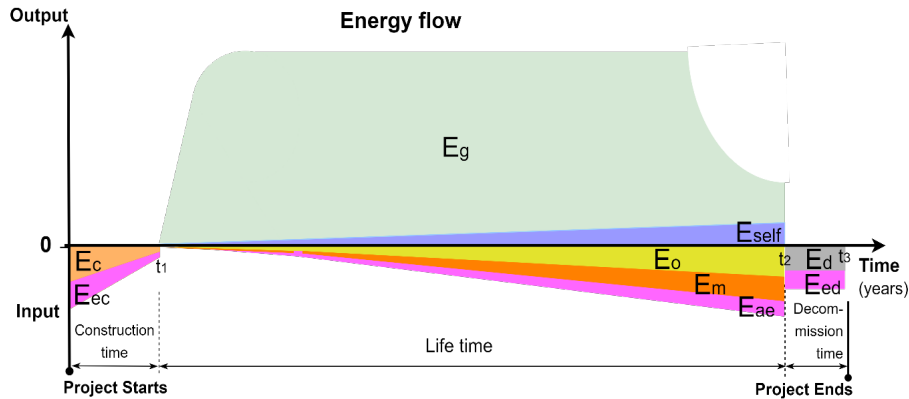


Figure 1. Energy flow for oil wells—energy inputs and outputs for oil extraction (not to scale)

E_g is gross energy, energy gains, energy output, produced energy or energy returned, E_c is energy required to construct facilities for energy production, E_{ec} is embodied energy required for construction of facility, E_o is energy used for run operations, direct energy for oil extraction, E_m is energy used for maintenance and repairs, E_{ae} is additional embodied energy used for operations, E_{self} is self-consuming energy for extracting oil, E_d is energy for decommission of objects after life time ends, E_{ed} is embodied energy required for decommission.

Embodied energy refers to the cumulative energy demand associated with the production of a material, component, or system, encompassing all upstream processes within the life cycle up to the point of delivery. This includes the energy expended in raw material extraction, refining, processing, manufacturing, assembly, and transportation. It represents the energy investment embedded in a product prior to its use phase. Weighty, embodied energy does not include the energy used during the product's operational life and end-of-life energy associated with disposal, recycling, or reuse.

The life cycle of an energy project typically consists of three main phases: the project construction phase ($t_{con} =$

t_1), the operational phase or life-time period ($t_{life} = t_2 - t_1$, i.e., the period during which the facility generates energy), and the decommissioning phase ($t_{decom} = t_3 - t_2$). These phases collectively define the total duration of the project ($t_{project} = t_3$). A key metric in evaluating the viability of such projects is the net energy (E_{net}), which represents the amount of energy available for societal use after accounting for energy expenditures across the system.

For this analysis, energy invested prior to the project (e.g., in infrastructure or earlier developments) is excluded due to the complexity of accurately quantifying its contribution. When the cumulative energy requirements for operation, maintenance, and embodied energy exceed the gross energy output, the extraction process becomes energetically and economically nonviable. In some systems, part or all of the operational and maintenance energy requirements may be met by the energy produced on-site, i.e., self-consuming energy.

When focusing specifically on oil extraction (i.e., the operation of oil wells), systems can be categorized into three distinct groups based on the source and method of energy investment (Figure 2). In the first category (a), all energy required for extraction processes is externally supplied, meaning no energy is sourced from the extracted oil itself. In the second category (b), the energy required for oil extraction is supplied through a combination of external sources and self-consumed energy derived from the extracted oil itself. In this case, a portion of the produced energy is reinvested into the extraction process, reducing dependency on external inputs. In the third category (c), all the energy required for oil extraction is supplied internally through self-consumption of the extracted oil. The system is energetically autonomous, with no need for external energy inputs.

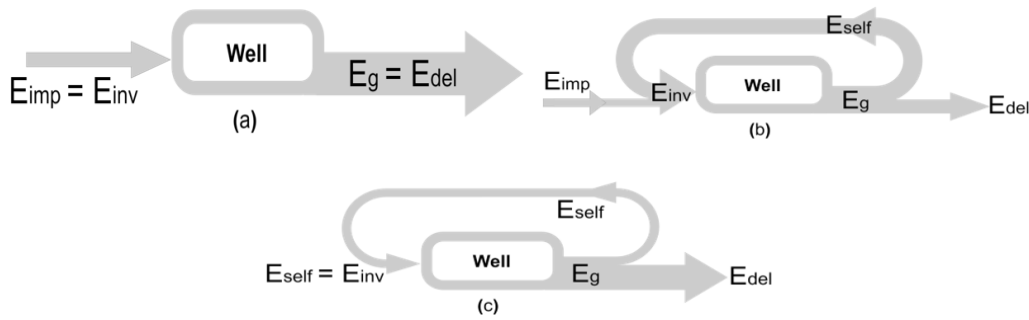


Figure 2. Oil wells energy flow scenarios: (a) using only imported energy for operations; (b) using imported and self-consumed energy for operations; (c) using only self-consumed energy for operations

The following relations are valid for the energy calculations (category a):

$$EROI_a = \frac{E_g}{E_{inv}} = \frac{E_{del}}{E_{imp}} = \frac{E_{net} + E_{imp}}{E_{imp}} = \frac{E_{net}}{E_{imp}} + 1 \quad (1)$$

$$E_{net} = E_g - E_{inv} = E_{del} - E_{imp} \quad (2)$$

where, E_{net} is net energy from the oil well, E_g is gross energy extracted from the oil well, E_{del} is oil energy delivered to society, E_{inv} is energy invested, and E_{imp} is imported energy for the operations of the oil well.

$$EROI_b = \frac{E_g}{E_{inv}} = \frac{E_{del} + E_{self}}{E_{imp} + E_{self}} \quad (3)$$

$$E_{net} = E_{del} - E_{inv} = E_g - E_{self} - E_{imp} \quad (4)$$

$$EROI_c = \frac{E_g}{E_{inv}} = \frac{E_{del} + E_{self}}{E_{self}} = \frac{E_{del} + E_{self}}{E_{self}} \quad (5)$$

$$E_{net} = E_g - E_{inv} = E_{del} - E_{self} \quad (6)$$

As documented in the studies (Guay-Boutet & Dufour, 2024; Heun & de Wit, 2012), the quantity of money (QM) a well can earn is:

$$QM_{oil} = P_{oil} \cdot E_{del} [\text{\$}] \quad (7)$$

where, QM_{oil} denotes the revenue from oil sales over a given period [USD], P_{oil} is the average market oil price during a given period [USD/barrel], and E_{del} is the quantity of oil sold [barrels]. The same revenue can alternatively be expressed as:

$$QM_{oil} = m_{up} \cdot C_{oil} \cdot E_g \quad [\$] \quad (8)$$

where, m_{up} is the mark-up ratio above production costs for the selling price of oil in the market during a given period [-], C_{oil} is production cost for extracting one barrel of oil in USD per barrel [\$/barrel], E_g is gross quantity of oil extracted from the well in barrels of oil [barrel]. From these two equations, we can derive the following:

$$QM_{oil} = P_{oil} \cdot E_{del} = m_{up} \cdot C_{oil} \cdot E_g \quad (9)$$

$$P_{oil} \cdot (E_g - E_{self} + E_{imp}) = m_{up} \cdot C_{oil} \cdot E_g \quad (10)$$

$$P_{oil} = \frac{m_{up} \cdot C_{oil}}{(1 - 1/EROI) + (E_{imp} / E_g)} \quad (11)$$

If we focus solely on oil wells from category (c), where all the energy required for extraction is self-consumed from the extracted oil and $E_{imp} = 0$ (Heun & de Wit, 2012), we can derive the following equation:

$$P_{oil} = \frac{m_{up} \cdot C_{oil}}{(1 - 1/EROI)} \quad (12)$$

Applying the formula for EROI, we get the following:

$$EROI = \frac{E_g}{E_{inv}} = \frac{V_{oil} \cdot e_{ints}}{QM_{inv} \cdot e_{inv}} \quad (13)$$

where, V_{oil} is the volumetric quantity of extracted oil in barrels [barrel], c_{ints} presents the energy intensity of extracted oil in Joules per barrel [J/barrel], c_{inv} represents the energy intensity of investment in Joules per USD [J/\$], and QM_{inv} is the QM invested in a particular oil well in USD [\$] during a given period.

Money return on investment, or MROI for an oil well, according to the studies (Guay-Boutet & Dufour, 2024; Heun & de Wit, 2012; King & Hall, 2011), can be defined as:

$$MROI = \frac{QM_{out} [\$]}{QM_{inv} [\$]} = \frac{V_{oil} \cdot P_{oil}}{QM_{inv}} \quad (14)$$

where, QM_{out} is revenue generated from a particular oil well in USD [\$] during a given period. The EROI value can then be calculated as follows:

$$\begin{aligned} EROI &= \frac{1}{QM_{inv}} \cdot \frac{V_{oil} \cdot e_{ints}}{e_{inv}} = \frac{MROI}{V_{oil} \cdot P_{oil}} \cdot \frac{V_{oil} \cdot e_{ints}}{e_{inv}} \\ EROI &= \frac{MROI}{P_{oil}} \cdot \frac{e_{ints}}{e_{inv}} \end{aligned} \quad (15)$$

and the price of oil (P_{oil}) is then:

$$P_{oil} = \frac{MROI}{EROI} \cdot \frac{e_{ints}}{e_{inv}} \quad (16)$$

3. Economic Value of US Oil in Comparative Perspective

3.1 Oil Value Relative to Gold

The value of oil can be assessed through a price-based comparison with gold. If the ratio between the price of an ounce of gold and the price of a barrel of oil is observed over the past 50 years, a clear long-term trend emerges (Figure 3).

If the ratio between the price of an ounce of gold and the price of a barrel of oil is observed over the past 50 years, the resulting trend can be illustrated in a diagram. The value of oil is assessed through two complementary approaches: a price-based comparison with gold and a comparison in energetic terms relative to manual labor. If the ratio between the price of an ounce of gold and the price of a barrel of oil is observed over the past 50 years, the resulting trend can be illustrated in a diagram (Figure 3).

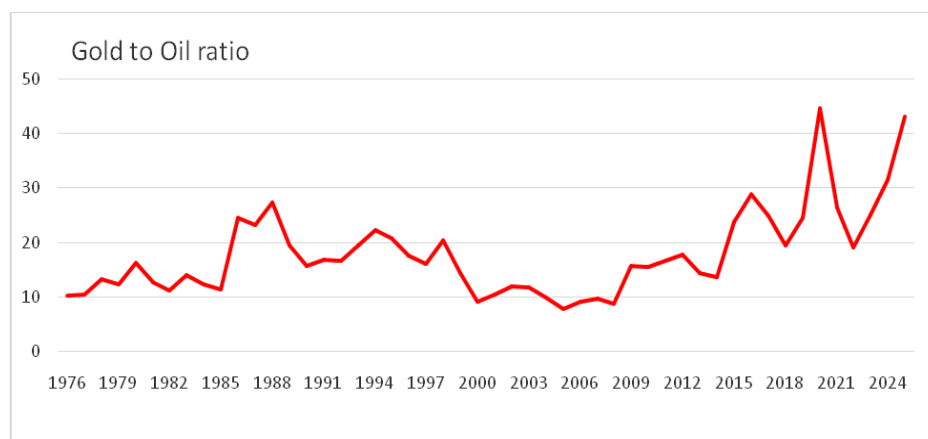


Figure 3. Gold prices to West Texas Intermediate (WTI) crude oil prices ratio over 50 years

Over this period, the average gold-to-oil ratio was 17.8, although it varied substantially over time. In the first decade (1976–1985), the ratio ranged between 10 and 12. It increased during 1986–1995 to values between 15 and 24, then declined during 1996–2005, averaging 12.9, with an absolute minimum of 7.86. From 2006 to 2015, the average ratio rose slightly to 14.5, while the average for the entire 1976–2015 period was 15.1. A sharp increase occurred after 2016, indicating a significant decline in the value of oil relative to gold. In 2020, the ratio surged to 44.7, reaching a record high of 53.65 in April 2025.

Although global oil reserves are declining and extraction is becoming increasingly challenging, the value of oil relative to gold has fallen markedly compared with previous decades. This suggests that oil has remained relatively inexpensive in recent years. However, such undervaluation is unlikely to persist indefinitely, and a future upward correction in oil prices relative to gold is plausible.

The long-term rise in the gold-to-oil ratio indicates a structural decline in oil’s purchasing power relative to scarce monetary assets. Despite increasing extraction complexity and depletion of conventional reserves, oil remains historically cheap compared with gold, suggesting a possible market mispricing of energy value.

3.2 Oil Value Relative to Human Labor

Human manual labor can also be interpreted in energetic terms. The realistic mechanical work output of a healthy human is approximately 0.6–0.8 kWh per day, assuming 8 hours of continuous manual labor, corresponding to an average mechanical power output of 75–100 W.

A barrel of crude oil contains approximately 1,700 kWh of thermal energy. Assuming a conversion efficiency of 30% in internal combustion engines, this yields about 510 kWh of usable mechanical energy. Given an average daily human mechanical output of 0.7 kWh, one barrel of oil is energetically equivalent to approximately 5,800 hours of human physical labor. With advanced engines reaching efficiencies above 40% internal combustion engine (Dorić et al., 2025), this increases to more than 7,700 hours of equivalent labor.

Over the past 50 years, the average ratio between wages for 5,800 hours of US labor and the price of one barrel of West Texas Intermediate (WTI) crude oil was approximately 2,391, fluctuating between 1,046 and 5,231 (Figure 4). In early 2025, this ratio was around 2,600, slightly above the long-term average.

This indicates that manual labor has been valued roughly 2,391 times more than crude oil, even when both are compared on the basis of equivalent mechanical energy output. While wages also reflect education, skills, and labor-market conditions, the comparison highlights the historically low economic valuation of oil-derived energy.

In Europe, this ratio is even higher, indicating an even greater valuation of labor relative to oil-derived energy.

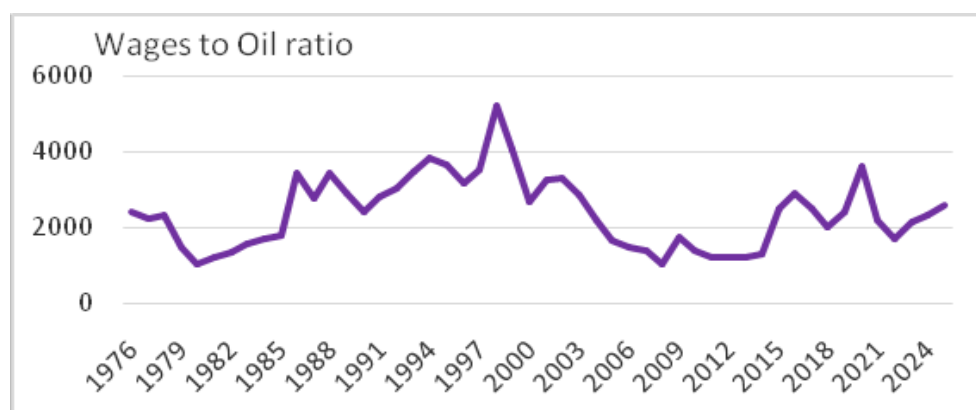


Figure 4. US labour wages to West Texas Intermediate (WTI) crude oil prices ratio over 50 years

Oil provides the equivalent work of thousands of human labor hours at extremely low cost, making it one of the cheapest productive inputs in modern economies. This exceptionally low valuation of concentrated energy has historically enabled high productivity growth and rising labor incomes.

3.3 Oil Productivity and Economic Output

Another measure of oil value is the GDP-to-oil ratio, defined as GDP per capita divided by annual oil consumption per capita (USD per barrel).

In 1976, this ratio was nearly identical for the US and the world, at 298 and 294, respectively (Figure 5). It increased steadily over time, and by 2008, both values again converged at approximately 2,084. Following the global financial crisis, however, divergence emerged. By 2023, the US ratio had risen to around 4,000, while the global average remained just above 2,900, indicating greater US economic output per barrel consumed.

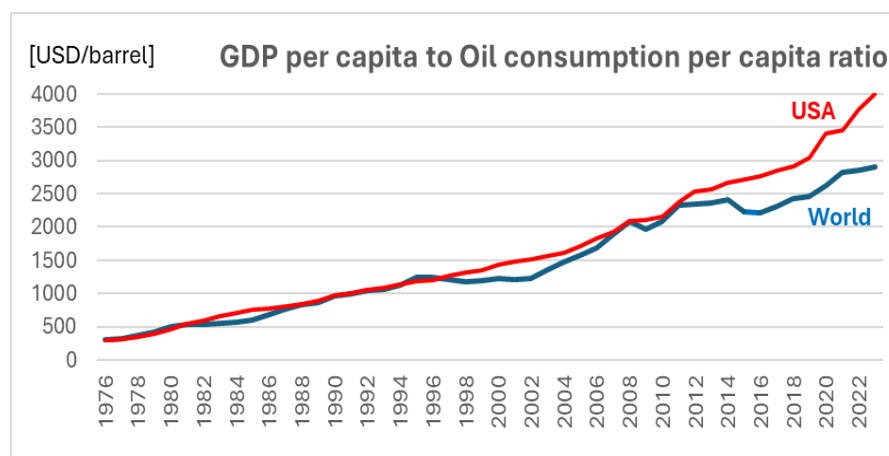


Figure 5. Trend of GDP per capita and oil consumption per capita ratio in the USA and globally over a period of 50 years

A similar pattern is observed in relative GDP and oil consumption per capita (Figure 6). In 1976, the US-to-world ratios for GDP per capita and oil consumption per capita were nearly identical (5.42 and 5.49, respectively). By 2023, the oil consumption ratio had declined modestly to 4.56, while the GDP per capita ratio increased to 6.28. This indicates that the US has reduced its relative oil intensity while simultaneously expanding its economic lead.

A strong historical correlation exists between GDP growth and fossil fuel use, particularly oil. Crude oil remains both a primary energy source and a strategic industrial feedstock. Its historically low price has significantly supported economic growth by lowering energy costs and stimulating production and consumption.

The steady rise in GDP generated per barrel indicates improving economic efficiency of oil use, especially in advanced economies. However, this efficiency still rests on access to abundant, relatively inexpensive oil. If oil becomes structurally more expensive due to depletion and declining EROI, significant macroeconomic

consequences may follow.

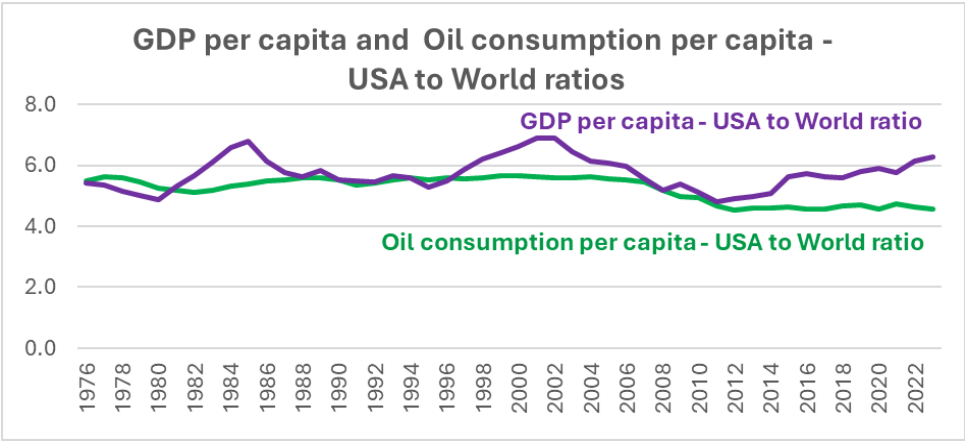


Figure 6. Trends in GDP per capita and oil consumption per capita ratios in the USA and globally over 50 years

Crude oil has been historically undervalued relative to monetary assets (gold), human labor, and the economic output it enables. This undervaluation has been a major hidden driver of industrial growth, high wages, and rising GDP. However, as oil extraction becomes increasingly energy-intensive and EROI declines, maintaining this economic model may become progressively more difficult.

4. Energy Return on Investment Trends of US Oil

Previous analytical studies consistently show a long-term decline in US oil EROI. Cleveland (2005) demonstrated that the net energy gained from US oil and gas extraction has decreased substantially over time due to rising energetic costs associated with increasingly difficult resource extraction. This paper projects both a linear and an exponential decline in EROI for US oil. Under the linear-decline scenario, the EROI of US oil would fall below 1 between 2030 and 2040, implying that oil extraction would no longer be economically viable. In the second scenario, the decline would be more moderate. Gagnon et al. (2009) confirmed a similar declining trend at the global level, while Heun & de Wit (2012) showed that declining EROI is closely linked to upward pressure on oil prices and to broader energy transitions. These findings indicate that declining EROI is a fundamental long-term constraint on both energy availability and economic development.

An analysis of the structure and volume of oil production in the USA reveals a significant increase post-2008, coinciding with the onset of the shale oil revolution (Figure 7). In 2005, almost all oil produced in the USA was conventional, with approximately 67% sourced from onshore and 33% from offshore. By 2015, more than half of US oil production was derived from shale, while conventional oil accounted for 48%, split between onshore and offshore sources.

Figure 7 illustrates the historical structure of oil production in the US, including shale, conventional, onshore, offshore, and Alaska production. The diagram shows three distinct peaks in US oil production, occurring in 1970, 1985, and more recently, with the lowest point recorded in 2008.

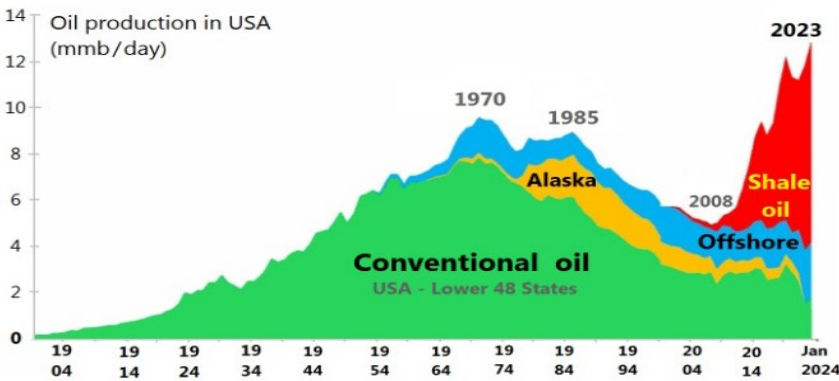


Figure 7. Oil production structure in the United States of America

In 2023, of the 12.9 million barrels per day of average US oil production, approximately 64.5% was shale oil, while only 35.5% was conventional oil from onshore and offshore sources, with oil production in Alaska declining toward negligible levels. These data from the US EIA (2025) are presented in Table 1.

Table 1. Average US oil production in the 21st century in millions of barrels of oil per day [mmb/day]

US Oil Production	2000	2005	2010	2015	2023
Total US oil production	5.82	5.18	5.476	9.42	12.90
Onshore oil production	4.05	3.47	3.154	2.66	2.27
Offshore oil production	1.77	1.71	1.732	1.87	2.31
Shale oil production	~0	~0	0.59	4.89	8.32
Onshore oil share [%]	69.59	66.94	57.60	28.24	17.60
Offshore oil share [%]	30.41	33.06	31.63	19.85	17.91
Shale oil share [%]	~0	~0	10.77	51.91	64.50

The “Red Queen” effect refers to the relatively short lifespan of hydraulically fractured shale oil wells, which tend to reach peak production earlier than conventional oil wells. Nearly three-quarters of a fracked shale oil well’s initial production is typically lost within the first 12 months, with a further half lost in the following 12 months. Operators must continuously bring new wells into operation at a sufficient rate to maintain overall shale oil production, this is known as the “Red Queen” effect (Figure 8).

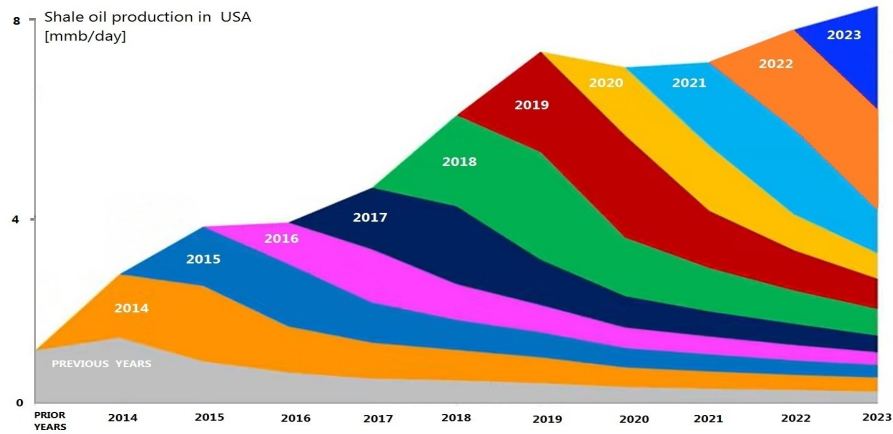


Figure 8. Shale oil Red Queen effect scheme

As a result, the EROI for shale oil declines rapidly within the first year of production, nearing its minimum values by the end of the second year of exploitation. The fracking process has been enhanced through horizontal drilling. However, these improvements accelerate further well depletion and cannibalization among adjacent wells.

As Cleveland (2005) provided data prediction on the decline of EROI for conventional oil, these values are included in Table 2. Although, in principle, the EROI for onshore oil is higher than that for offshore oil, it was not possible to find separate data; therefore, identical average values based on the estimates and the study (Heun & de Wit, 2012) were used. We can include the weighted EROI calculation:

$$EROI_{weighted} (US\ oil) = \sum EROI_j \cdot x_j \quad (17)$$

where, $EROI_j$ is the estimated EROI value for oil type (conventional or shale), and x_j is the share in production.

Table 2. US oil energy return on investment (EROI) estimate according to decline trend and oil structure

Weighted US Oil EROI	2000	2005	2010	2015	2023
Conventional oil $EROI_{min}$	10.8	9.4	8.0	6.7	4.3
Conventional oil $EROI_{max}$	12.5	11.6	10.8	10.1	9.0
Shale oil $EROI_{min}$	3	3	3	3	3
Shale oil $EROI_{max}$	6	6	6	6	6
Weighted US oil $EROI_{min}$	10.80	9.40	7.46	4.78	3.46
Weighted US oil $EROI_{max}$	12.50	11.60	10.28	7.97	7.07

Relations between net energy from oil (E_{net}) shown in Table 3, and US produced oil (E_g):

$$E_{net} = E_g \cdot (1 - 1 / EROI) \quad (18)$$

Table 3. Net energy from US oil (million barrels/day) according to the decline trend and oil structure (see Table 2)

Net Energy from US Oil [mmb/day]	2000	2005	2010	2015	2023
Total US oil production	5.82	5.178	5.476	9.42	12.9
Min net energy from US oil	5.28	4.63	4.74	7.45	9.17
Max net energy from US oil	5.35	4.73	4.94	8.24	11.07
Min net energy share from US oil [%]	90.74	89.36	86.60	79.08	71.11
Max net energy share from US oil [%]	92.00	91.38	90.28	87.46	85.85

If the EROI of oil decreases from 20 to 10, the resulting reduction in net energy is only about 5%. However, a decline from 10 to 2 leads to a net energy loss of up to 40%. For a global daily oil production of about 100 million barrels in 2023, with oil accounting for roughly 30% of the total energy mix, a one-point decline in the global oil EROI results in a substantial net energy deficit. This deficit must be compensated either by increased oil production or by replacement with alternative energy sources, which becomes increasingly costly.

5. Estimation of Recent US and Russian Oil Energy Return on Investment

Eqs. (7) to (16) define the relationships between the market price of oil (P_{oil}), its production/extraction costs (C_{oil}), EROI, and MROI. Using these equations and data on oil prices and costs, an EROI estimate is derived for oil produced in the US and Russia. US oil is a mix of conventional and shale oil, while Russian oil consists entirely of conventional crude from onshore and offshore sources.

It is difficult to determine the actual average production cost of oil at the country level. Table 4 provides data (Statista, 2015) on the cost of producing a barrel of oil (C_{oil}) for 2015 in the world's top oil-producing countries. These figures are based on data from more than 15,000 oil fields across 20 countries, compiled in November 2015.

Table 4. Oil average production costs— C_{oil} [in USD per barrel of oil] in 2015 worldwide

Oil Production Costs C_{oil} [2015 USD/barrel]	UK	Canada	USA	Norway	Russia	Iran	UAE	KSA	Kuwait
Total oil cost	52.5	41	36.3	36.1	17.3	12.6	12.3	9.9	8.5
Operating cost	30.7	22.4	14.8	12.1	8.4	5.7	5.7	5.4	4.8
Capital cost	21.8	18.7	21.5	24	8.9	6.9	6.6	4.5	3.7

The year 2015 experienced relatively stable oil prices, making it suitable for analyzing the relationship between EROI and oil production across countries. An examination of oil prices shows that the average closing price of Brent crude was USD 53.03, exceeding the total production cost of UK oil in 2015. This suggests that oil extraction from the North Sea was already approaching unprofitability, reflecting a low EROI for UK oil. The study (Hamilton, 2024) suggests that by 2031, the EROI of UK oil will fall below one, rendering it energy-negative. Saudi Arabia (KSA) and Kuwait have the lowest oil production costs globally, positioning them favorably in the market and allowing for high MROI. This also indicates that these countries likely have the highest EROI among major oil producers.

The initial analysis focuses on oil parameters in the US, Russia, and Saudi Arabia (see Figure 9). In 2015, the price of WTI crude oil averaged USD 48.66, the OPEC reference basket USD 49.49, and the Ural blend USD 51.23. In 2023, the price of WTI crude oil averaged USD 77.76 (see Figure 10), while the average price of the Ural blend remained USD 51.23.

The mark-up factor in the US has been declining for decades and, according to the study (Heun & de Wit, 2012), was approximately 1.2 in 2010. By applying Eq. (16), together with the methodology outlined by the studies (Guay-Boutet & Dufour, 2024; Heun & de Wit, 2012; King & Hall, 2011), the resulting diagram (Figure 9) reflects a US oil mark-up ratio of 1.2 applied to all three countries. Figure 9 clearly shows that the EROI for oil is higher in Russia, and especially in Saudi Arabia (KSA), than in the US. This indicates that the breakeven price of oil is significantly lower in Russia and Saudi Arabia. The ratio of oil production costs reflects not only EROI but also MROI in these countries. Consequently, the actual mark-up for Russian and Saudi oil in 2015 was substantially higher than the assumed value of 1.2 shown in Figure 9.

The average production cost for US oil in 2023 was approximately USD 62 per barrel. For this cost level, Figure 10 shows that the breakeven price of US oil is much higher than in 2015, reaching around USD 80 for EROI values above 10, and increasing sharply as the average EROI declines further. The mark-up value in the US for 2023 is

likely lower than 1.2 and certainly not higher. This indicates an even more unfavorable relationship between the breakeven price of US oil and its EROI.

Reliable data on oil production costs for Russia and Saudi Arabia in 2023 are not publicly available; therefore, Figure 10 does not include breakeven prices for these countries.

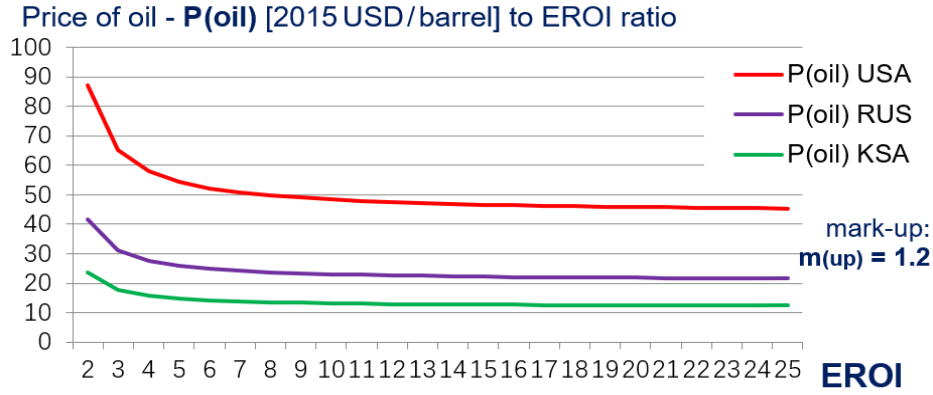


Figure 9. Oil price (P_{oil}) to energy return on investment (EROI) ratio for 3 main oil-producing countries in 2015, with an equal mark-up ratio value of 1.2

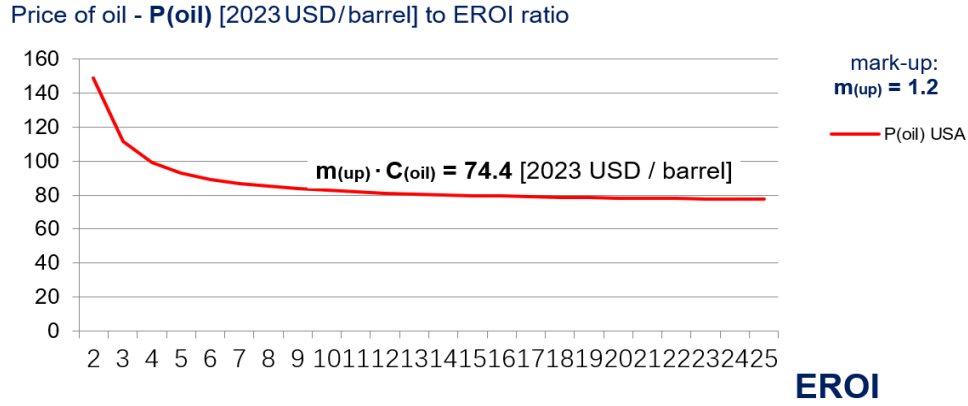


Figure 10. Oil price (P_{oil}) to energy return on investment (EROI) ratio for the USA in 2023, with mark-up ratio value of 1.2

Safronov & Sokolov (2014) calculated the hydrocarbon EROI for the five largest Russian oil companies. Their results show that in 2012 the EROI for Russian oil and gas ranged from 21 to 35. By applying the same weighting method (Eq. (17)) used for US production, based on production volumes in 2012, the weighted average EROI for Russian oil and gas is estimated as $EROI_{rus}(2012) = 25.4$. The actual EROI is likely lower, as the analysis does not include smaller producers and less favorable fields. Additionally, it is well established that EROI for natural gas is generally higher than for oil, further suggesting that the EROI for Russian oil alone in 2012 was below 25. Assuming an EROI value of 25 for Russian hydrocarbons in 2015, the Russian oil MROI for 2015 can be calculated as follows:

$$\begin{aligned}
 MROI_{rus}^{2015} &= P_{oil}(rus) \cdot EROI_{rus}^{2015} \cdot \frac{e_{inv}}{e_{ints}} = \\
 &= 51.23[2015\$/barrel] \cdot 25 \cdot \frac{9[MJ/\$]}{6100[MJ/barrel]} = 1.89
 \end{aligned} \tag{19}$$

The energy investment for Russian oil exploration, denoted as $e_{inv}(rus) = 9$ [MJ/\$] is assumed to be lower than that for US oil, where the energy investment was calculated in the range of $e_{inv}(usa) = 14-33$ [MJ/\$], same as the studies (Gagnon et al., 2009; King & Hall, 2011). If a higher value for energy investment in Russian oil production is assumed, the resulting MROI would be even greater for the same EROI.

Based on the calculated oil MROI of 1.89 for Russia and the assumed oil MROI of 1.1 for the USA, as used by

the studies (Guay-Boutet & Dufour, 2024; King & Hall, 2011) and considering different values of invested energy (e_{inv}), we will calculate the assumed oil EROI values for both the USA and Russia. By using Eq. (20), we present the results of average EROI values for two large oil producers in tabular form (Table 5 and Table 7).

$$EROI_{year} = \frac{MROI}{P_{oil}(year)} \cdot \frac{e_{ints}}{e_{inv}(year)} \quad (20)$$

Table 5. Assumed weighted energy return on investment (EROI) values for US oil

Year	Oil Price [\$/Barrel]	MROI	EROI		
			$e_{inv} = 14$ [MJ/\$]	$e_{inv} = 19$ [MJ/\$]	$e_{inv} = 33$ [MJ/\$]
2005	WTI: 56.64	1.1	8.46	6.24	3.59
2010	WTI: 79.48	1.1	6.03	4.44	2.56
2015	WTI: 48.66	1.1	9.85	7.26	4.18
2023	WTI: 77.76	1.1	6.17	4.54	2.62
2005	WTI: 56.64	1.2	9.23	6.80	3.92
2010	WTI: 79.48	1.2	6.58	4.85	2.79
2015	WTI: 48.66	1.2	10.75	7.92	4.56
2023	WTI: 77.76	1.2	6.72	4.96	2.85

Note: MROI = Monetary return on investment; WTI = West Texas Intermediate, e_{inv} = energy investment for oil exploration

The assumed EROI values for US oil in 2015 and 2023 (Table 5) are consistent with the results presented in Section 3. In the previous analysis (Table 2), EROI values for US oil in 2015 range from 4.78 to 7.97, while for 2023 they range from 3.46 to 7.07. In this section, the estimated EROI values for 2015 range from 4.18 to 9.85, and for 2023 from 2.62 to 6.17, depending on the assumed energy investment. Although it is unlikely that the MROI for US oil is as high as 1.2, these results are also reported in Table 5. Net energy from US oil (E_{net}) calculated using Eq. (18) and the EROI values in Table 5, is presented in Table 6.

Table 6. Net energy estimate from US oil according to Eq. (20) and monetary return on investment (MROI) = 1.1 (data from Table 5)

Net Energy From US Oil [mmb/day]	2010	2015	2023
Total US oil production	5.476	9.42	12.9
Min net energy from US oil	3.34	7.17	7.98
Max net energy from US oil	4.57	8.46	10.81
Min net energy share from US oil [%]	60.94	76.08	61.83
Max net energy share from US oil [%]	83.42	89.85	83.79

An analysis of EROI values for US oil in 2000 and 2023 shows a significant decline. Based on the average values obtained from both estimation methods, the EROI decreased from 11.60 in 2000 to 4.83 in 2023. As a result, the net energy loss from US oil exceeds 12% per barrel over this period.

By applying Eq. (20), the EROI for Russian oil in 2023 can be estimated in the range from 16.64 to 20.34 (Table 7) assuming the same MROI of 1.89 as in 2015.

Table 7. Assumed weighted energy return on investment (EROI) values for Russian oil

Year	Oil Price [\$/Barrel]	MROI	EROI		
			$e_{inv} = 9$ [MJ/\$]	$e_{inv} = 11$ [MJ/\$]	$e_{inv} = 14$ [MJ/\$]
2023	Ural: 62.99	1.89	20.34	16.64	13.07

Note: MROI = Monetary return on investment; e_{inv} = energy investment for oil exploration

Russia's actual MROI in 2023 is likely lower than this estimate, both due to the previously mentioned factors and because discounted oil prices were offered in response to sanctions related to the conflict in Ukraine. Therefore, Figure 11 presents results for lower assumed MROI values for Russian oil in 2023. Precise estimation of recent EROI values for Russian oil remains challenging due to the lack of reliable data on current production costs.

For the equal amount of invested energy ($e_{inv} = 14$ MJ/\$) in oil extraction, Russian oil exhibits more than twice the EROI of US oil, as shown in Table 5 and Table 7. This difference is partly due to variations in MROI values, among other factors mentioned.

Estimating EROI values for other major oil-producing countries (such as Saudi Arabia, Kuwait, and the UAE) is not feasible due to the lack of well-level EROI data and reliable information on production costs and energy investment per barrel. However, based on 2015 production costs and oil market prices, it can be concluded that

EROI and MROI in Gulf countries are higher than in Russia and significantly higher than in the US.

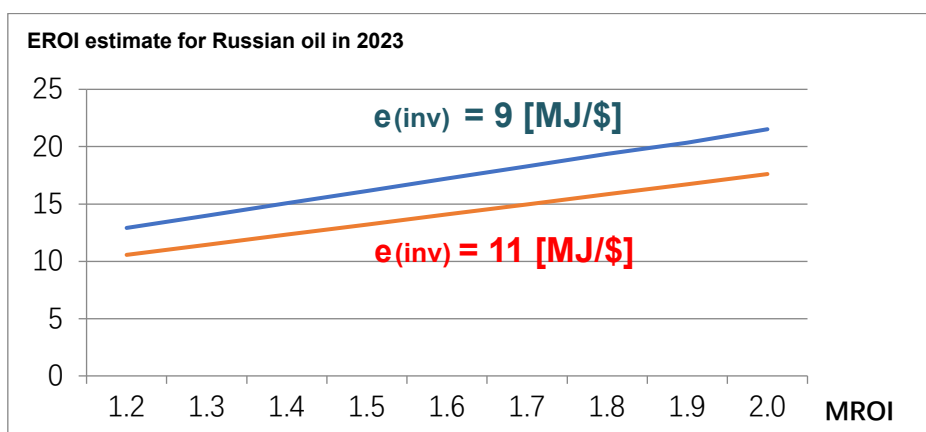


Figure 11. Calculated energy return on investment (EROI) values for Russian oil in 2023 for different monetary return on investment (MROI) values

6. Overview of Recent US Oil Production

Oil production in the US is closely linked to the number of active drilling rigs. Figure 12 (top) presents the number of oil and gas rigs operating in the US from 1973 to 2025, based on annual averages, while Figure 12 (bottom) shows the average monthly number of active oil and gas rigs in the US after 2000.

In the early 1980s, the number of active oil and gas drilling rigs in the US exceeded 4,500. During the first 15 years of the 21st century, the annual average ranged between 1,000 and 1,900 rigs, with a peak of 2,017 recorded in October 2011. Following this peak, the number of active rigs declined significantly, reaching a low of just 250 in April 2020 (Figure 12, bottom).

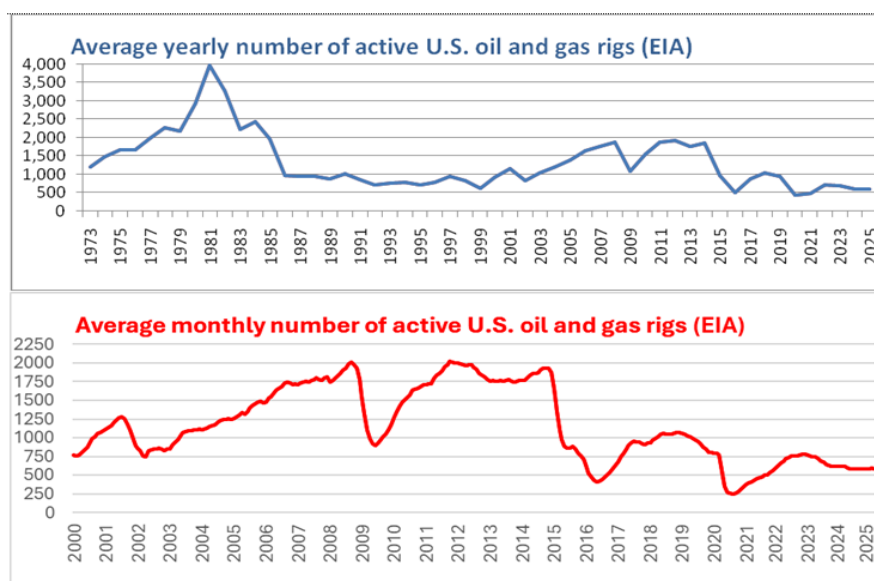


Figure 12. Average number of active US oil and gas rigs

Note: EIA = US Energy Information Administration

In recent years, the distribution of active rigs in the US has followed an approximate ratio of 82:17:1 for oil, gas, and miscellaneous rigs, respectively. As of the end of April 2025, there were 483 active oil rigs, 99 gas rigs, and 5 miscellaneous rigs.

The overall reduction in rig activity primarily affects shale oil operations, which are more responsive to market fluctuations. In contrast, conventional onshore and offshore platforms tend to remain relatively stable, as they are not started up or shut down as frequently.

US oil production declined to an 11-month low in January 2025, primarily due to a reduction in shale oil output. This trend is particularly evident in the Permian Basin, reflecting the fast-maturing nature of many shale formations.

These shale plays are characterized by high initial production rates that decline rapidly (the “Red Queen” effect), underscoring the challenges of sustaining long-term output. Although US oil production exceeded 13 million barrels per day at the beginning of 2025, a further decline is anticipated under current conditions. This is primarily due to the reduction in the number of active drilling rigs and the economic unviability of many oil fields at current market prices.

7. Conclusion

This paper indicates that the EROI for oil in the US is currently relatively low, with limited evidence suggesting a near-term increase; instead, a gradual global decline in EROI appears likely. As the global population continues to grow, alongside increasing per capita energy consumption, the importance of energy systems becomes increasingly critical. This trend necessitates the development of additional energy infrastructure and greater allocation of resources for energy production, posing a significant global challenge. Energy is neither created by technology nor by financial capital; it is derived from material resources, which are finite and largely non-renewable. Modern society often underestimates the fundamental role of energy in sustaining economic activity and shaping future development.

The global decline in EROI for oil and gas, particularly in the US, the world’s largest oil producer, leads to a reduction in the net energy available for societal use, requiring compensation through alternative energy sources. This trend is likely to influence not only the US but also the global energy system, potentially increasing the strategic importance of regions such as Russia and the Gulf countries. Global oil production may face constraints in the coming years, while natural gas production could approach a peak toward the end of this decade before stabilizing or gradually declining. Together, these developments point to increasing pressures on the global energy system.

The G7 countries, including the largest EU economies, consume approximately 3.4 times more energy per capita than the rest of the world. With fossil fuels accounting for around 70% of energy consumption in the EU, rapid substitution with solar and wind energy remains challenging. In 2022, solar energy accounted for about 2% of global primary energy use, wind approximately 3%, and nuclear energy less than 4%. A transition from low shares of renewables to substantially higher levels within a few decades would be technically and economically demanding and would require significant infrastructure development. Petroleum continues to serve as a fundamental energy base for modern society, while alternative infrastructures to fully support renewable energy systems are still evolving. Without substantial technological and systemic advancements, combined with a continued decline in oil EROI, the global socio-economic system may face increasing adjustment pressures.

In the near term, the European Union may face energy-related challenges due to reduced oil and gas imports from Russia, as well as declining electricity generation from aging nuclear power plants. Over the next 5 to 10 years, this could contribute to slower economic growth and potential impacts on living standards, driven by energy constraints, rising prices, and resource limitations. To address these challenges, the EU will need to prioritize improvements in energy efficiency and conservation, while adapting societal practices toward reduced energy consumption.

Author Contributions

Conceptualization, S.M., N.P., B.N., and D.M.; methodology, S.M. and B.N.; validation, S.M. and B.N.; formal analysis, S.M.; investigation, S.M. and N.P.; data curation, S.M.; writing—original draft preparation, S.M.; writing—review and editing, S.M., N.P., and D.M.; visualization, S.M. and N.P.; supervision, S.M. and D.M. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflict of interest.

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